

# Milan George

## Gameplay and Systems Programmer | Unreal Engine 5, C++

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### SUMMARY

Gameplay and systems programmer with six years in game development, four of them across real-time strategy games and VR training simulators. Currently leading development of a Total War style strategy title, architecting large scale unit simulation on Unreal Engine 5. Comfortable owning a domain end to end: architecture, implementation, and directing the engineers building it.

### TECHNICAL SKILLS

**Languages:** C++, C#, C

**Engines:** Unreal Engine 5, Unity

**Unreal:** Mass Entity (ECS and data oriented design), plugin and module architecture, UMG and Slate, Common UI, navigation and pathfinding, vertex animation for crowd rendering, Niagara, FMOD integration, Blueprint exposure and tooling

**Architecture:** Data oriented design, ECS, MVVM and data binding, event driven systems, MVC, Factory, State, Observer, Object Pool, Service Locator, Command

**Tools:** Git, Plastic SCM, Visual Studio, Rider, Unreal Insights

### PROFESSIONAL EXPERIENCE

#### Project Lead, Nytwolf Games

Bangalore, India | April 2024 to Present

Project Greenleaf (Unreal Engine 5, PC). A Total War style strategy game with large scale real-time battles and a campaign overworld.

- Lead development across both the **battle and overworld layers**: architecture, systems design, implementation, task breakdown, and code review for a team of five.
- Architected soldier simulation on Unreal **Mass Entity** as a hybrid model, where Mass drives per soldier simulation and movement while a paired actor handles animation and rendering. Sustained **4,000 individually simulated soldiers** in a single battle, using vertex animation to keep crowd rendering affordable.
- Designed and shipped the battle system as a **standalone Unreal plugin**, decoupled from the campaign layer, independently testable, and retained as reusable in-house technology for future titles.
- Built a **dynamic UI framework**: an MVVM style data binding layer where widgets subscribe to named fields on shared data models and update reactively, so gameplay code never holds widget references. Type dispatch binding supports any widget class, and the full API is Blueprint exposed so designers wire interfaces without engineering support.
- Designed and implemented an **A\* grid navigation and pathing layer** for unit movement and formation handling.
- Built a **settlement designer and economy layer** with stronghold style interior building placement, modular wall construction, and an Anno 1800 style production chain economy, cut in a later design pivot.
- Profiled and optimised the game with **Unreal Insights**, tracing frame time and memory bottlenecks across the simulation, navigation, and rendering paths and resolving them.

#### VR Gameplay Programmer, i3 Simulations

Bangalore, India | August 2022 to April 2024

- **SimGym** (Unreal Engine, PC VR): built a scenario creator mode with a fully functional multigrid system, letting users author detailed simulations without engineering support.
- **ResusVR** (Unity): automated the localization pipeline, cutting turnaround time and cost. Designed and implemented medical training scenarios across varied situations and age groups.

- **ENT Surgery VR** (Unity): engineered an ffmpeg based session recorder with multi session management for surgical simulations, built an analytics and instructor grading system, and significantly improved load times.

### **Technical Interviewer and Code Reviewer, Outscal**

*Remote, part time | October 2022 to August 2023*

- Conducted technical interviews and code reviews for game development candidates, assessing engineering skill and maintaining code quality standards.

### **Unity Developer (Intern), StudioKrew**

*May 2021 to August 2021*

- Built synchronous multiplayer prototypes with Photon, a 3D Whack-A-Mole game and a four player golf game, implementing core gameplay and networked systems.

### **SELECTED PROJECT**

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#### **Benji 2D, a 2D game engine built from scratch (C++)**

*[github.com/mln-grg/Benji-2D](https://github.com/mln-grg/Benji-2D)*

- Multi platform 2D framework built on a signature based ECS with packed component pools, a frame rate independent game loop, a type indexed event bus, AABB collision handling, and a Dear ImGui debug layer.

### **EDUCATION**

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**B.Tech, Computer Science Engineering.** IHRD Kallopparra, APJ Abdul Kalam Technological University | 2022

**Full Stack Game Development.** Outscal | 2020 to 2022